



Deployment & Use OTAinfo HAL

Not to be used in production

Version 1.0.3

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URL: www.otainfo.com

Scope

The document is for the non production version of OTAinfo HAL (Hardware Abstraction Layer) for Seeed Studio XIAO nRF52840 Sense on Zephyr. OTAinfo HAL will abstract (a) Sensors: Gyroscope, Accelerometer & Temperature sensor and (b) Communication: Bluetooth. A pull mechanism is used to get current reading from sensors. The pull is initiated by the Bluetooth connected device. All sensor readings are exposed as one Bluetooth service.

Goal

Provide an easy to use experience for IoT solution development companies. Using OTAinfo, companies can reduce porting time for new devices from months to minutes.

[Proceed to next page]

Content of package

File	Description
Seeed_studio_xiao_nRF52840_sense.zip	Contains the HAL -required object and configuration files.
main.c	Sample code to learn how to use OTAINFO HAL
hal_02_seed_studio_xiao_nrf52840_sense.c.obj	Object file that needs to be linked during build process
hal_02_seed_studio_xiao_nrf52840_sense.h	GSMAAdapter's header file
CMakeLists.txt	Sample to demonstrate how to include the OTAINFO file(s)
prj.conf	Sample provided
sample.yaml	Sample provided

[Proceed to next page]

Environment

Zephyr

Version file:

```
VERSION_MAJOR = 3  
VERSION_MINOR = 6  
PATCHLEVEL = 99  
VERSION_TWEAK = 0  
EXTRAVERSION =
```

Commit:

commit 859cc2f1feec93132e06753278be02889418333f (HEAD -> main, origin/main, origin/HEAD)

Author: Johann Fischer <johann.fischer@nordicsemi.no>

Date: Mon Dec 4 11:43:29 2023 +0100

Or later

West

West version: v1.2.0 or later

Character Encoding

UTF-8

Note: UTF-8 format must be used on all receivers.

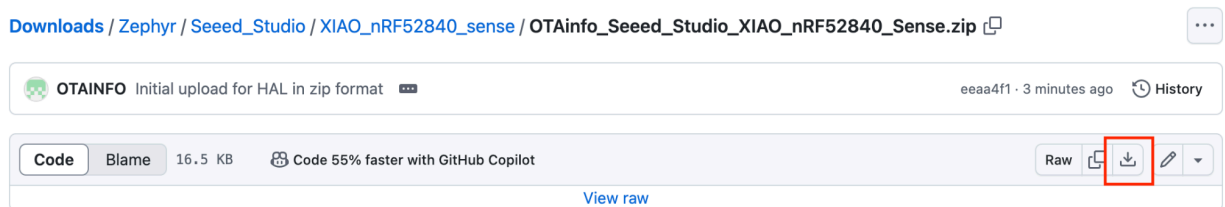
How to install Zephyr?

Follow the installation instructions on the Zephyr [Installation page](#).

How to Download OTAINfo HAL?

The HAL can be downloaded as a zip file: [OTAInfo_seeed_Studio_XIAO_nRF52840_Sense.zip](#).

Click the download button on the provided link.



Where do I compile and deploy?

1. Unzip the folder to extract the **OTAInfo-02-001** directory (complete contents shown in Appendix B).
2. Copy this folder inside Zephyr (it can be placed as a sibling to the **samples** folder in the Zephyr deployment).
3. If necessary, you can rename **OTAInfo-02-001**.

From the CLI inside Zephyr, run:

```
west build -b xiao_ble/nrf52840/sense [folder name | OTAInfo-02-001]
```

The build will generate a **zephyr.uf2** file at **build/zephyr/**.

To Deploy:

1. Plug the XIAO nRF52840 board into the computer.

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2. Double-click the RST button to set the device into UF2 bootloader mode.
3. Open the Finder window; the XIAO nRF52840 will show up as a USB drive.
4. Drag and drop the `zephyr.uf2` file onto this drive.
5. The XIAO nRF52840 will deploy and reset. Deployment is complete.

How do I know service has started?

- Use a serial monitor to connect to the USB port where the XIAO nRF52840 is connected. It should display a message confirming the service has started.

```
----- Closed serial port /dev/tty.usbmodem14201 due to disconnection from the machine -----
---- Opened the serial port /dev/tty.usbmodem14201 ----
*** Booting Zephyr OS build v3.6.0-3947-g859cc2f1feec ***

OTAINfo, Inc. grants rights for non commercial use only. Build for:

    Board: Seeed Studio XIAO nRF52840 Sense
    RTOS: Zephyr
    ver: t.dp.1.10
Begin test for ST lsm6xxx

Starting Bluetooth peripheral

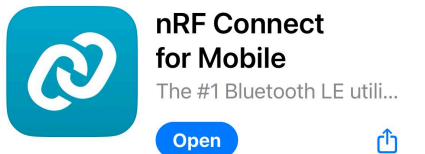
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[00:00:00.342,346] <inf> bt_hci_core: hci_vs_init: HW Platform: Nordic Semiconductor (0x0002)
[00:00:00.342,376] <inf> bt_hci_core: hci_vs_init: HW Variant: nRF52x (0x0002)
[00:00:00.342,407] <inf> bt_hci_core: hci_vs_init: Firmware: Standard Bluetooth controller (0x00) Version 3.6 Build 9
[00:00:00.343,261] <inf> bt_hci_core: bt_dev_show_info: Identity: DE:40:48:40:43:3C (random)
[00:00:00.343,292] <inf> bt_hci_core: bt_dev_show_info: HCI: version 5.4 (0x0d) revision 0x0000, manufacturer 0x05f1
[00:00:00.343,322] <inf> bt_hci_core: bt_dev_show_info: LMP: version 5.4 (0x0d) subver 0xffff
Bluetooth initialized
```

- The Bluetooth client can connect, and data from the sensors is fetched on demand.
- Use the nRF Connect mobile app to fetch data from the XIAO nRF52840.

Instructions for nRF Connect:

1. Install the app on your mobile device.
2. Open the app; the scanner will detect nearby Bluetooth devices.
3. Connect with `OTAINfo999`.
4. To view the data, you may need to change the format to UTF-8.



132 RATINGS	AGE	CATEGORY	DEVI
4.2 ★★★★☆	4+ Years Old	Utilities	Nordic

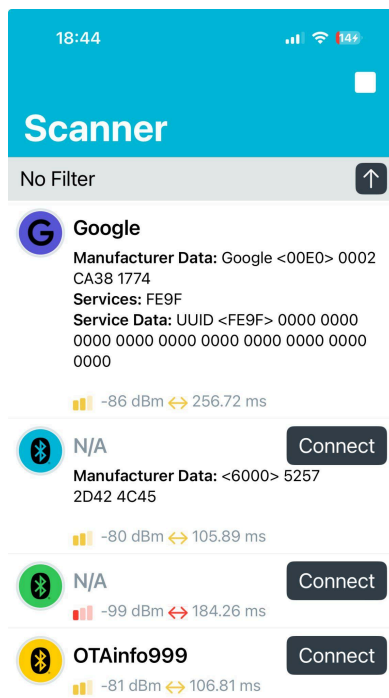
What's New [Version History](#)

Version 2.7.11 1w ago

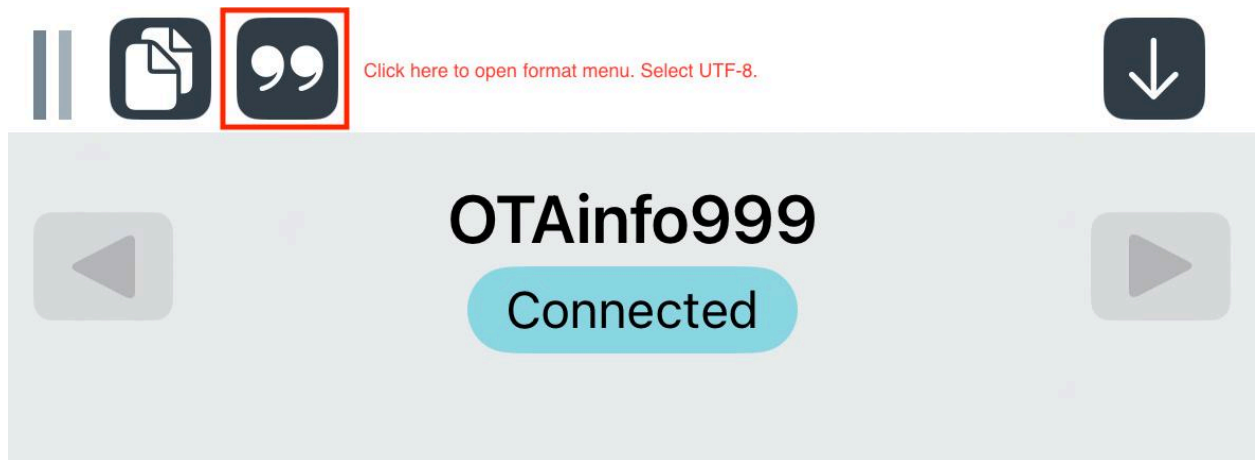
Fixes for a couple of crashes, one of them related to DFU. The other one to Peripheral Logs.

Install the app on your cell phone.

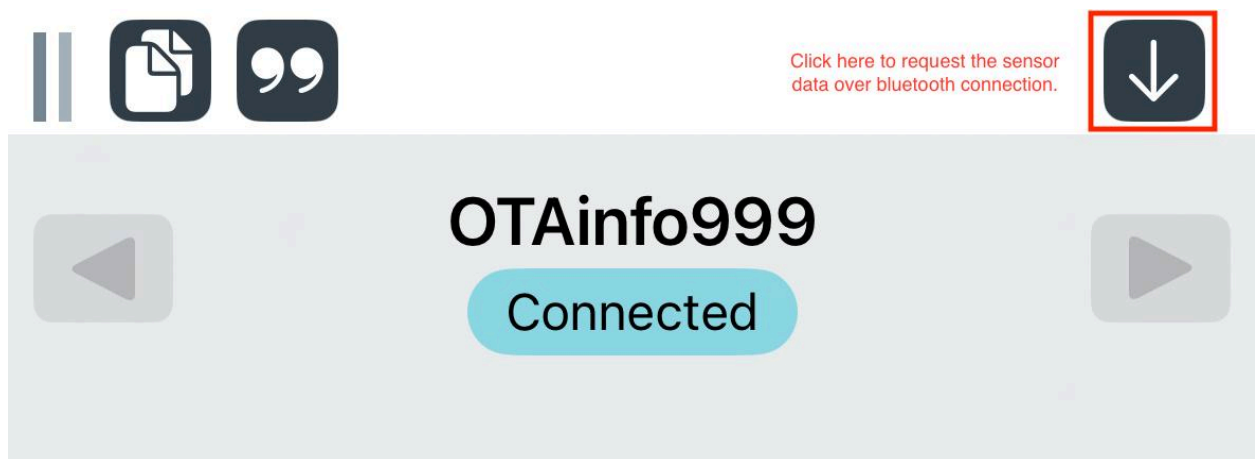
On opening the app, the scanner will pick up nearby Bluetooth devices. Connect with OTAINfo999.



To view the data, the user might have to change the format to UTF-8. Watch the [video](#) for better understanding.



(this image is magnified)



(this image is magnified)

If you have any questions, feel free to contact us.

Appendix A

CMakeLists.txt

```
cmake_minimum_required(VERSION 3.20.0)
find_package(Zephyr REQUIRED HINTS $ENV{ZEPHYR_BASE})
project(<project-name>)
FILE(GLOB app_sources src/*.c)
target_sources(app PRIVATE ${app_sources})
target_include_directories(app PRIVATE ${CMAKE_CURRENT_SOURCE_DIR}/include)
target_link_libraries(app PRIVATE
${CMAKE_CURRENT_SOURCE_DIR}/lib/hal_02_seed_studio_xiao_nrf52840_sense.c.obj)
j)
```

Appendix B

```
<project-name/OTAinfo-02-001>
|
|__<include>
|    |__hal_02_seed_studio_xiao_nrf52840_sense.h
|
|__<lib>
|    |__hal_02_seed_studio_xiao_nrf52840_sense.c.obj
|
|__<src>
|    |__main.c
|
|__CMakeLists.txt
|
|__prj.conf
|
|__sample.yaml
```